Paisabazaar Credit Risk Dashboard: Project Documentation

Prepared by: Nitin Tiwari **Tool:** Tableau Public (data cleaning in Python / Google Colab) **Dashboard link:** https://public.tableau.com/app/profile/nitin.tiwari2142/viz/Paisabazaar_Dashboard/PaisabazaarCreditRiskDashboard

1. Project Title

Paisabazaar Credit Risk Dashboard: Understanding which customer behaviours are linked with poor credit scores.

2. Dataset Name and Description

Dataset: Paisabazaar customer credit dataset (the same dataset used in my Python EDA project).

- **Size:** 100,000 records and 28 columns in the raw file. After cleaning and feature creation, 30 columns.
- **Structure:** 12,500 unique customers observed over 8 months (Month 1 to 8), so each row is one customer-month record.
- **Content:** demographics (age, occupation), income and salary, bank accounts, credit cards, loans, outstanding debt, credit utilization, delayed payments, payment behaviour, credit mix, and the target variable **Credit Score** (Poor, Standard, Good).

3. Dashboard Objective

To turn my EDA findings into an interactive dashboard that answers one business question:

> *Which financial behaviours are most associated with a Poor credit score, and which customer groups may need closer risk review?*

The dashboard supports credit-risk assessment, customer segmentation, and risk-aware product recommendations.

4. Data Cleaning and Transformation

Cleaning was done in Python (Google Colab) before connecting the data to Tableau.

- Checked shape, data types, duplicates and null values (0 duplicates, 0 nulls).
- Removed sensitive and unnecessary columns (ID, Name, SSN) because Tableau Public data is publicly visible. Customer_ID was kept only to count distinct customers.
- Converted Age and count columns (accounts, cards, loans, delayed payments, inquiries, delay days) to integers.
- Treated 12,007 invalid "NM" values in Payment_of_Min_Amount as "Unknown" instead of forcing them into Yes or No.
- Created new columns for better visualization: Age_Group, Income_Band (Low, Mid, High), Utilization_Band, Delay_Band and Debt_to_Income.

- In Tableau, created a calculated field, % Poor Records, and used Count Distinct on Customer ID for the customer KPI.
- Cleaning notebook (Google Colab): https://colab.research.google.com/drive/1JSGl9AvqmK-RAn6yQFET8LckZHvJHuQL?usp=sharing

5. Dashboard Overview

The dashboard has 7 visuals, 3 interactive filters and a key-insights panel.

Visual	Type	Why this chart
Total Customers	KPI	Shows the size of the customer base (12,500 unique customers)
% Poor Credit Records	KPI	Shows overall risk level at a glance
Avg Credit Utilization (%)	KPI	Shows how much of available credit customers use on average
Credit Score Split	Pie chart	Shows each score category as a share of the whole
Poor % by Income Band	Bar chart	Compares the Poor share across income groups on a fair, percentage basis
Avg Delayed Payments by Score	Bar chart	Compares one average across the three score groups
Poor % by Min Payment	Bar chart	Compares the Poor share across Payment of Min Amount categories

Filters: Occupation, Credit Mix and Age Group (applied to all charts and KPIs).

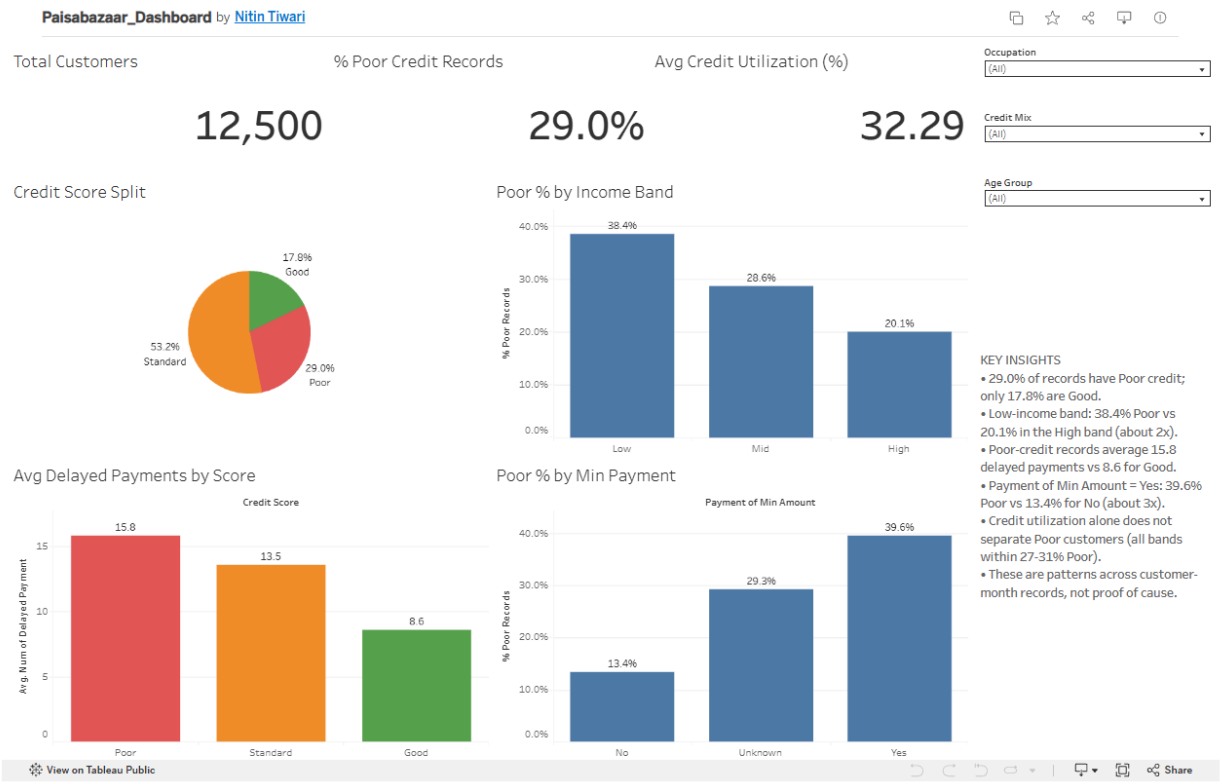


Figure 1: Paisabazaar Credit Risk Dashboard (full view).

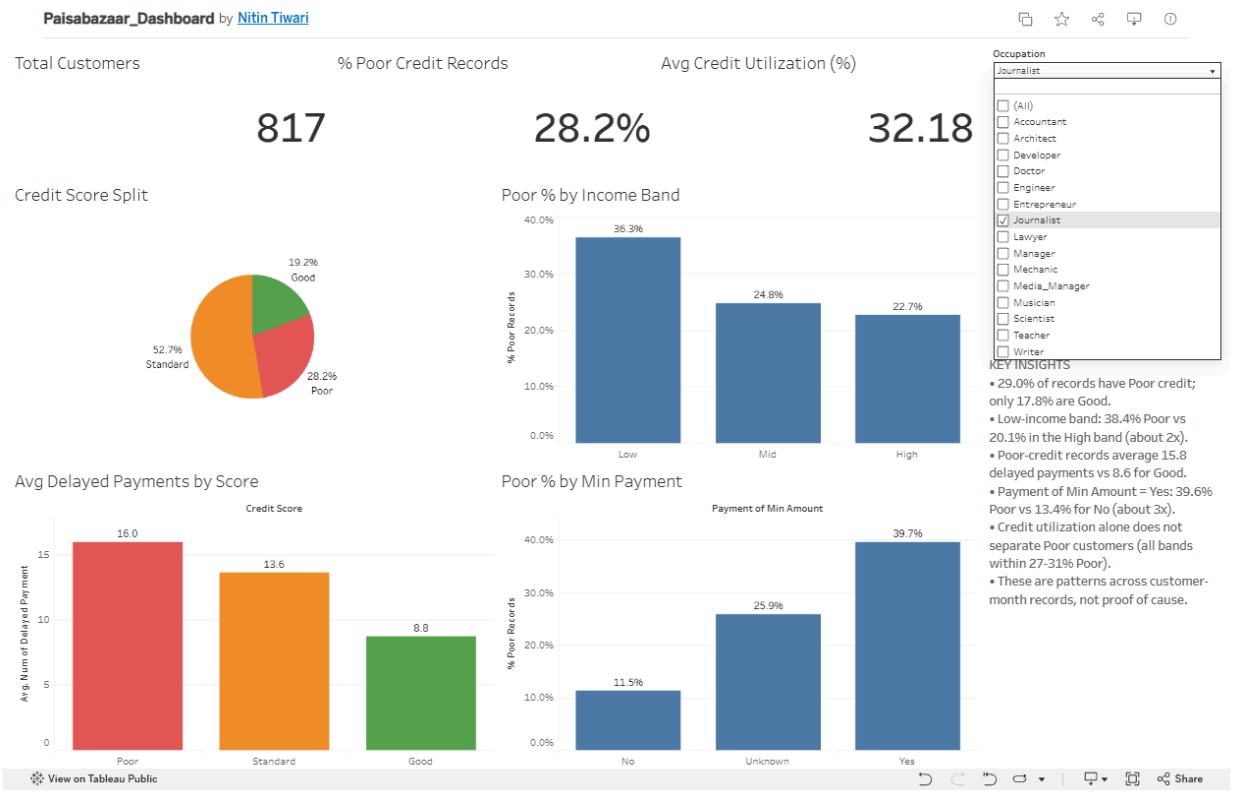


Figure 2: Dashboard with Occupation filter applied.

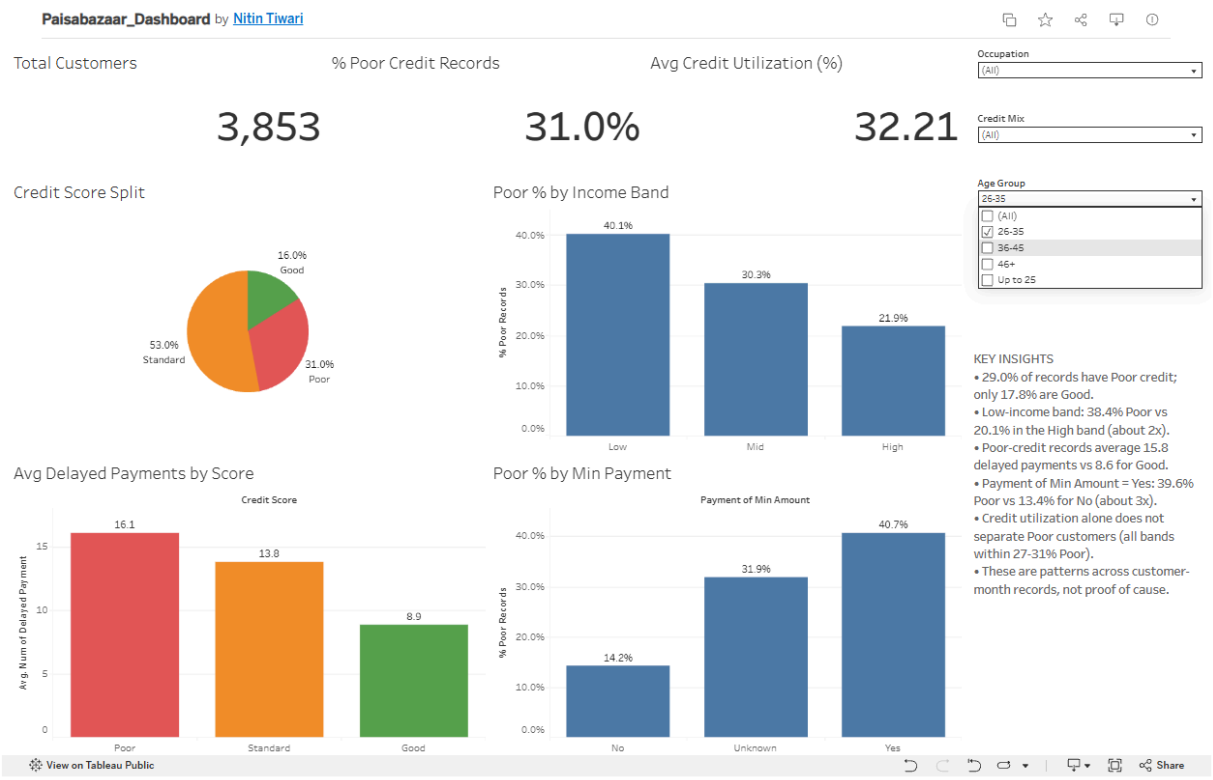


Figure 3: Dashboard with Age Group filter applied.

6. Key Insights

1. **29.0% of records have a Poor credit score**, 53.2% are Standard and only 17.8% are Good.
2. **Income matters:** the Low income band has 38.4% Poor records versus 20.1% in the High band, about 2 times higher.
3. **Late payments are a strong signal:** Poor-credit records average 15.8 delayed payments versus 8.6 for Good-credit records.
4. **Payment of Min Amount:** records marked Yes have 39.6% Poor versus 13.4% for No, about 3 times higher.
5. **Credit utilization alone does not separate Poor customers:** all four utilization bands fall between 27% and 31% Poor.
6. **Occupation does not drive risk:** across all 15 occupations the Poor share stays between 26.3% (Media_Manager) and 30.6% (Entrepreneur), close to the overall 29%.

7. Business Impact Summary

- **Risk monitoring:** delayed payments and the minimum-payment behaviour can be used as early-warning signals for customers who may need review.
- **Customer segmentation:** the Low income segment shows a higher Poor share, so it may benefit from extra repayment guidance or more cautious credit limits.

- **Smarter assessment:** because utilization and occupation do not separate risk on their own, credit decisions should combine several factors instead of relying on a single one.
- **Portfolio view:** with 29.0% Poor and 17.8% Good, a large part of the base sits in Poor or Standard, so risk management and guidance remain important.

8. Limitations

- Each row is a customer-month record, so percentages describe records, not individual customers.
- The patterns show association, not proof of cause.
- There is no confirmed fraud label, so the analysis points to risk patterns only.
- The exact definition of Payment of Min Amount should be confirmed against the dataset's data dictionary.

9. Next Steps

- Add a predictive model for the credit score category.
- Add anomaly detection to flag unusual customer profiles.
- Extend the dashboard with a high-risk customer watchlist.

10. Links

- **Tableau Public dashboard:** https://public.tableau.com/app/profile/nitin.tiwari2142/viz/Paisabazaar_Dashboard/PaisabazaarCreditRiskDashboard